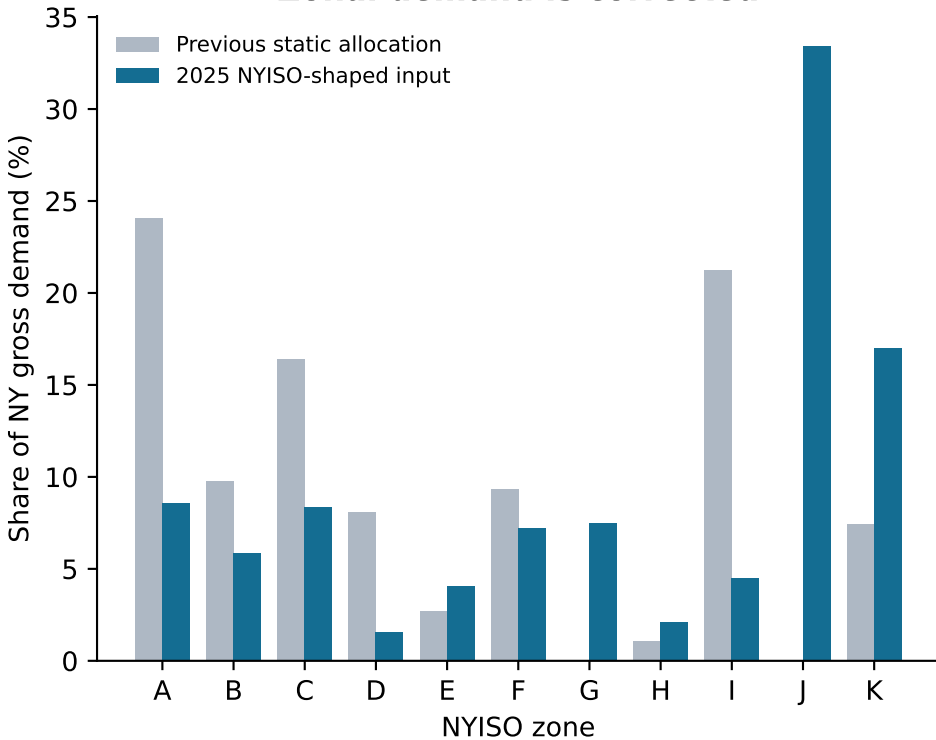
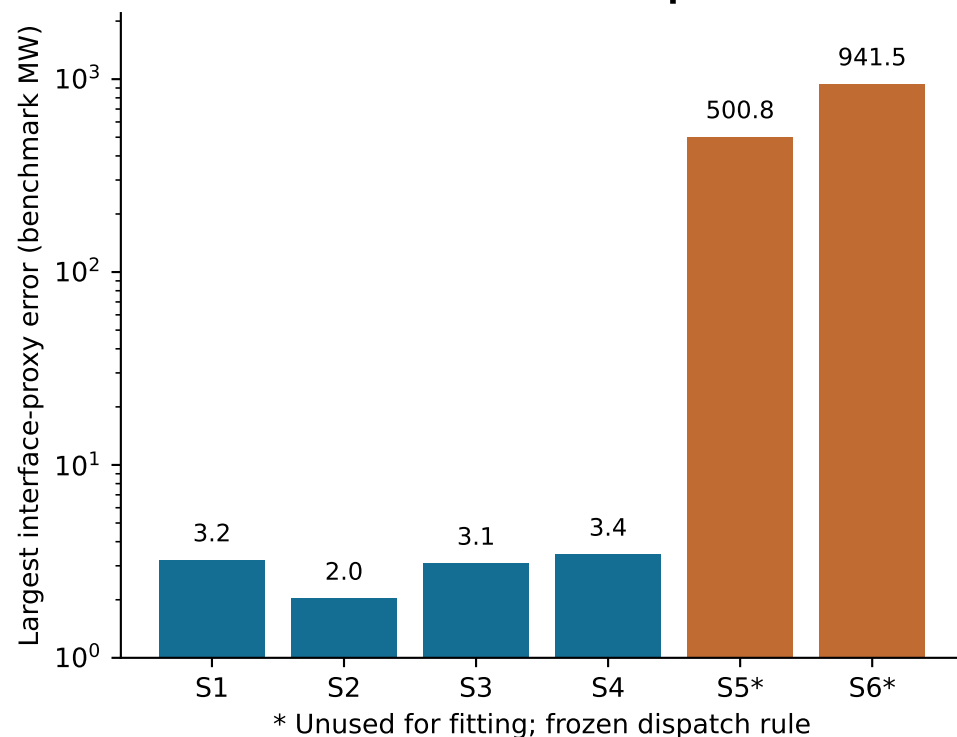


66-bus NPCC-derived New York research baseline

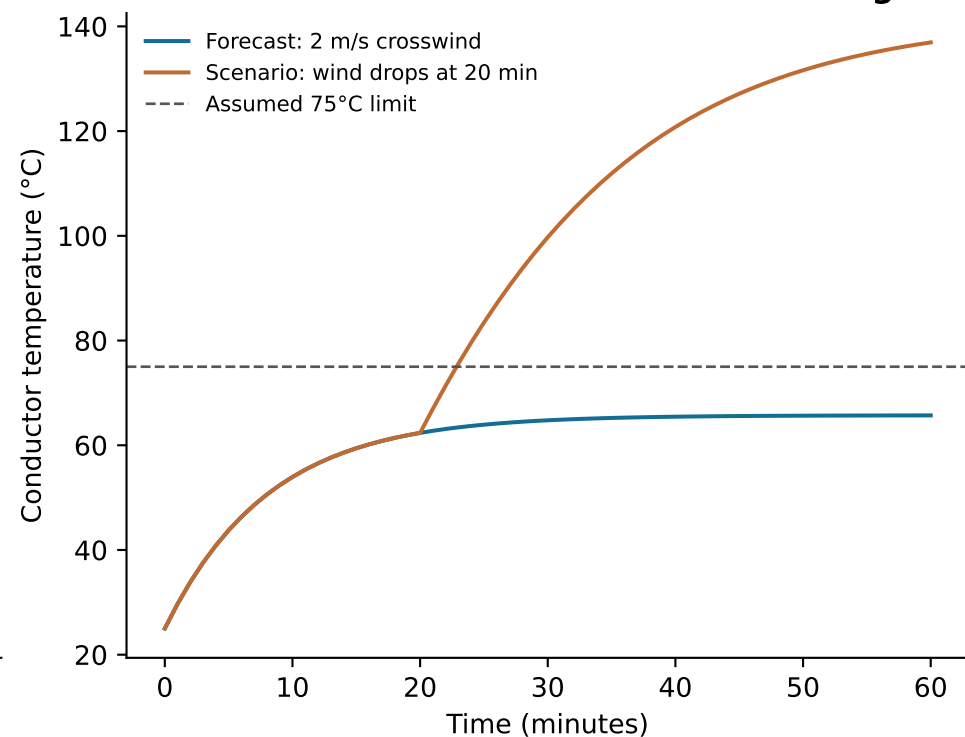
Zonal demand is corrected



Calibration differs from prediction



Weather errors can cause overheating



Selected 2025 summer benchmark = 10,902 MW. Interface operators and conductor realizations are approximations. Temperature trace uses prescribed current and synthetic weather; it is separate from the steady AC thermal solves.